superior blood glucose control in the complex hospital environment. **Methods**: We performed a retrospective review of electronic medical records from a single university hospital. The hospital implemented new order sets allowing prandial insulin dosing based on an ICR and inpatient nursing staff received necessary training. We compared glucose levels over the first three days after admission in hospitalized patients prescribed prandial insulin either as a fixed dose or based on an ICR. Patients on fixed dose insulin were selected from a time 3 months prior to the implementation of ICR order sets, to avoid any bias in patient selection. Patients on ICR dosing were selected from 3 months after implementation of ICR order sets, to allow adequate time for initial implementation. Our inclusion criteria included patients admitted to both medicine and surgery services, with Type 1 or Type 2 DM, age between 18-80. Exclusion criteria included transplant patients, patients on insulin infusion, pregnancy, steroid use and dialysis. 65 patients were included in each group. Outcome measures included average blood glucose over 72 hours, fasting and postprandial hypoglycemia (<70 mg/dL) and hyperglycemia incidence (>180 mg/dL). **Results:** Average glucose over 3 days was 167 +/- 39 mg/dL and 162 +/- 33 mg/dL and did not differ between groups. However, a higher percentage of glucose values were in target range (70-180 mg/dL) in the fixed dosing group (67.9%) compared to ICR (62.5%, p=0.018). The incidence of hypoglycemia was low and did not differ between groups (1.2% in both). However, patients had more overall hyperglycemia with ICR dosing (36% vs 31%, p=0.018), particularly pre-lunch hyperglycemia (52% vs. 38%, p=0.007). Fasting glucose was similar between groups.

Conclusions: In conclusion, our study demonstrated that prandial insulin administered based on ICR did not improve overall glycemic control or reduce incidence of hypoglycemia in hospitalized patients with diabetes. In fact, overall and post-meal rates of hyperglycemia were generally higher with ICR dosing. This could be related to inaccurate counting of carbohydrates or delayed timing of insulin administration as it was given after the food was consumed. Additional studies are needed to further evaluate the effectiveness of flexible meal dosing and the impact on patient satisfaction and staff workload.

Diabetes Mellitus and Glucose Metabolism

GESTATIONAL DIABETES, DIABETES IN PREGNANCY, AND IN UTERO EXPOSURES

Secretagogin Levels Are Unrelated to Gestational Diabetes Mellitus in a Cohort of Pregnant Women

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SUN-639

Introduction

Secretagogin (SCGN) is a calcium binding protein related to insulin release in the pancreas. Although SCGN is not co-released with insulin, plasma concentrations have been found to be increased in type 2 diabetes mellitus patients.[1,2,3] Up to this day, no study on SCGN levels in patients with gestational diabetes mellitus (GDM) has been published.

Patients and Methods

In 138 women of a high-risk population for GDM at the Medical University of Vienna, secretagogin levels of GDM patients were compared to women with a normal glucose tolerance (NGT). Glucose tolerance, insulin resistance and secretion were assessed with an oral glucose tolerance test (oGTT) performed before 20 weeks gestation. The women with GDM (39.1%) were further divided into GDM types depending on insulin sensitivity or secretion defects defined as below the 25th percentile in the oGTT of the NGT controls.

Results

Compared to women with NGT (mean SCGN= 52.7 ng/dl), there was no statistically significant difference in SCGN in patients with GDM (mean value 53.9 ng/dl, p=0.857). After splitting into secretion defect and insulin resistance subtypes, SCGN remained unrelated to GDM in our study population. However, Secretagogin was found to be significantly higher in postpartum visits (mean= 62.9 ng/dl) than during pregnancy (mean value= 48.5 ng/dl; p= 0.047). Furthermore, SCGN was positively correlated with BMI (p=0.006) in the present analysis.

Conclusion

Unlike in studies conducted on type 2 diabetes,[1,2,3] a relationship between GDM and Secretagogin levels could not be demonstrated in this study. However, lower levels during pregnancy point towards physiological changes in SCGN levels unrelated to (gestational) diabetes mellitus. Further research, ideally including before-pregnancy levels, is paramount to assess possible roles of SCGN during pregnancy.

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Adrenal

ADRENAL CASE REPORTS II

Spontaneous Adrenal Hemorrhage in a Patient with Antiphospholipid Syndrome on Rivaroxaban: A Potentially Fatal Complication.

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SUN-192

Introduction: Primary Adrenal insufficiency is an uncommon complication of antiphospholipid syndrome, with an incidence of 0.4% [2]. It is often secondary to bilateral adrenal hemorrhage. We present a case of bilateral adrenal hemorrhage in a patient with APLS on anticoagulation with rivaroxaban. **Case:** 46-year-old m with a history of antiphospholipid syndrome, recently transitioned from

warfarin to rivaroxaban for anticoagulation, who presented to the ED after a syncopal episode following a prior episode of abdominal pain with an unremarkable work-up. He subsequently developed severe fatigue, dizziness, headaches, nausea and 15 lbs weight loss. On presentation, the patient was hypotensive(72/45 mmHg) and tachycardic. Intravenous hydration was started with minimal response. Initial laboratory testing showed serum sodium of 121mmol/L, potassium of 5.5mmol/L and random cortisol of 0.8mcg/dL. The patient was admitted to the intensive care unit where he was started on vasopressors and hydrocortisone 50 mg IV every 8 hours. A non-contrast CT of the abdomen and pelvis showed thickening of the adrenal glands with decreased attenuation. MRI of the abdomen showed hyper-intensity of the adrenal glands bilaterally (T1 images), without post-contrast enhancement suggestive of bilateral adrenal hemorrhage. His electrolytes normalized, and he was successfully discharged home on hydrocortisone and fludrocortisone replacement with outpatient follow-up.
Discussion: Atraumatic bilateral adrenal hemorrhage is rare, but remains one of the most common endocrine-related complications of antiphospholipid syndrome (APLS). The venous anatomical configuration of the adrenal gland increases risk of thrombotic hemorrhagic infarction[1]. Patients with APLS are commonly anticoagulated to prevent thrombosis. The ideal anticoagulation regimen remains controversial. Only three other cases of spontaneous bilateral adrenal hemorrhage on patients with APLS using new oral anticoagulants (NOACs) were reported. The use of NOACs seem to increase the already-elevated risk of adrenal hemorrhage seen in patients with APLS.
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Thyroid

THYROID NEOPLASIA AND CANCER

Impact of Nodule Size on the Probability of Hurthle Cell Carcinoma and Other Cancers in Thyroid Nodules with Multiple Chromosomal Copy Number Alterations

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OR21-02

Fine-needle aspiration (FNA) of thyroid nodules yields indeterminate cytological diagnosis in ~20% of cases, confounding patient management. This includes Hurthle cell nodules, which typically yield Bethesda IV and III cytology. Chromosomal copy number alterations (CNA) are known to occur in thyroid tumors, particularly in Hurthle cell carcinomas (HCC) as well as in other typically follicular-patterned tumors including papillary thyroid carcinomas (PTC) and poorly differentiated thyroid carcinomas (PDTC). The aim of this study was to evaluate thyroid nodules tested positive for CNA but negative for all other genomic alterations using ThyroSeq v3 NGS assay in order to establish the probability of cancer in these nodules and find whether it is influenced by the pattern of CNA and nodule size. We evaluated 111 nodules with multiple CNA detected by ThyroSeq in FNA samples and available surgical pathology outcome. Of those, 69 (62%) nodules showed CNA changes consistent with genome near-haploidization (GNH) whereas 42 (38%) nodules had multiple chromosomal losses and gains (CLG). Nodule size ranged from 0.5-10.2 cm; cytology was Bethesda III in 54%, Bethesda IV in 43%, and Bethesda V-VI in 3% of cases, with Hurthle cells mentioned in the cytology report in 64% of cases. On surgical pathology, 38 (34%) of these nodules were malignant (including 24 HCC, 8 PTC, and 5 oncocytic PDTC) and 73 (66%) were benign (including 43 Hurthle cell and 18 follicular adenomas). No significant difference was observed in probability of malignancy between the two patterns of CNA (p=0.41). However, a significant correlation between the nodule size and probability of cancer was found (p=0.006). In specific CNA groups, correlation between cancer and nodule size remained significant for nodules with GNH pattern (P=0.0002), but not with CLG pattern (p=0.449). Specifically, cancer probability in nodules with GNH pattern and <2 cm in size was 14% (all cancers minimally-invasive), 2.0-2.9 cm was 33%, 3.0-3.9 cm was 50%, 4-4.9 cm was 67%, and ≥5 cm was 80%. Among high-risk cancers (widely-invasive or angioinvasive HCC, PDTC), all 10 tumors had the GNH pattern (p=0.01) and average nodule size of 4.9 cm (range, 2.1-8.5 cm). These findings suggest that CNA of both types are frequently found in Hurthle cell tumors, and probability of cancer in nodules with CNA and no other mutations increases with larger nodule size. This may help to refine the pre-operative assessment of cancer probability and risk of more aggressive disease and offer more tailored management to these patients.

Reproductive Endocrinology

CLINICAL STUDIES IN FEMALE REPRODUCTION I

Fluoxetine Administration Influences Serotonin-Driven Bone Remodeling During Lactation and Pregnancy Outcome in Mice

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SAT-020

Selective serotonin reuptake inhibitors (SSRI) are the most commonly prescribed class of antidepressants during pregnancy and lactation. SSRI decrease bone mineral density (BMD) across all ages and sexes. Lactation is also characterized by increased bone resorption to mobilize calcium due to the demands of lactation and achieves this via a serotonin-induced hormonal cascade. This serotonin-mediated bone loss is normally restored after weaning but is persistent when an SSRI is administered during